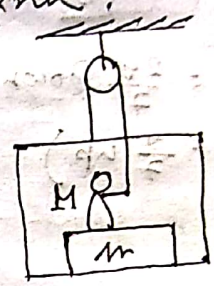
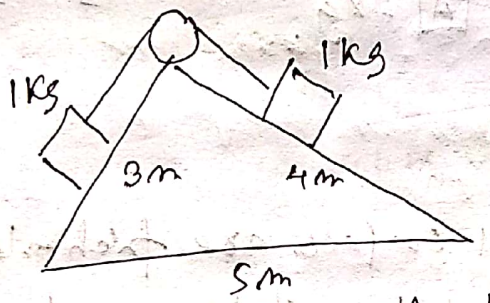
N.L.M (JEE) Problem.

(1) A man of mass 'M' is standing on a plank kept in a box. The plank and box as a whole has mass 'm'. A light string passing over a fixed smooth pulley connects the man and box. If the box remains stationary, find the tension in the string and the force exerted by the man on the plank.

$\left(\frac{(M+m)g}{2}, \frac{(M-m)g}{2}\right)$

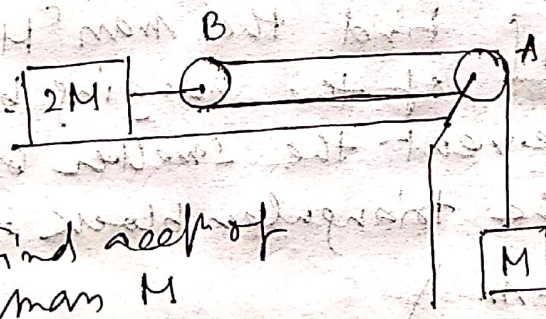


(2)



All surfaces are frictionless and the string and the pulley are light. Find magnitude of accelerations of two blocks. (g/10)

(3)



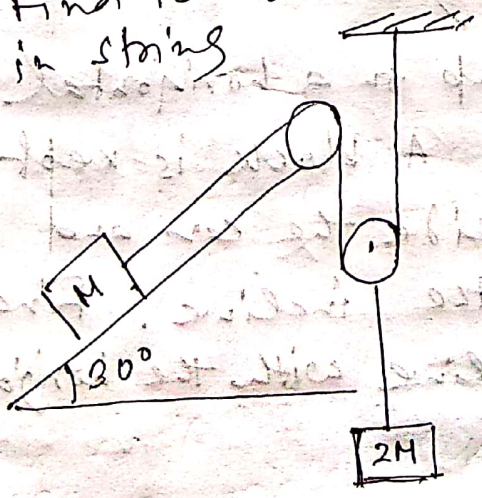
Both pulleys and the string are light and all surfaces are frictionless.

a) Find acc of mass M

b) Find tension in string

[Ans a) 2g/3
b) Mg/3]

(4)



All surfaces are frictionless and the pulleys and the string are light. Find acceleration of the block of mass 'M'.

$\left(\frac{g}{3} \text{ up the plane}\right)$

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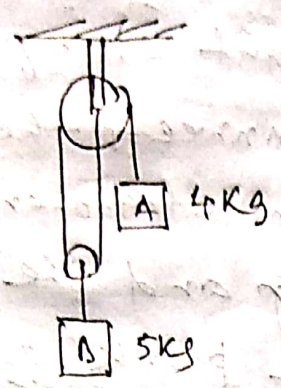
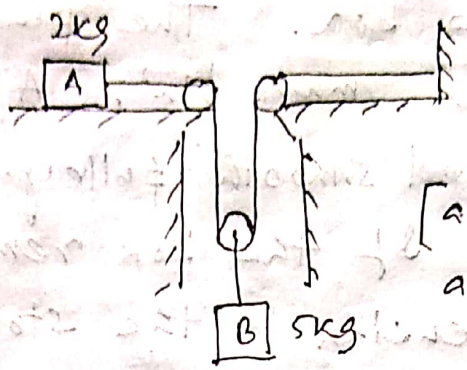
5. Find accl^n of blocks in the three situations shown in fig.

a)



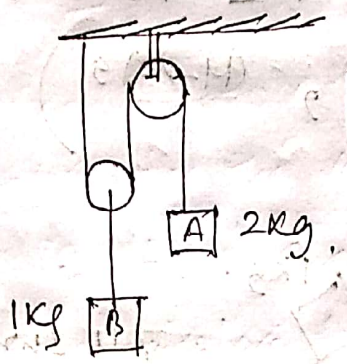
$\left(a_A = \frac{2g}{7} \text{ down}, \; a_B = \frac{g}{7} \text{ up}\right)$

b)



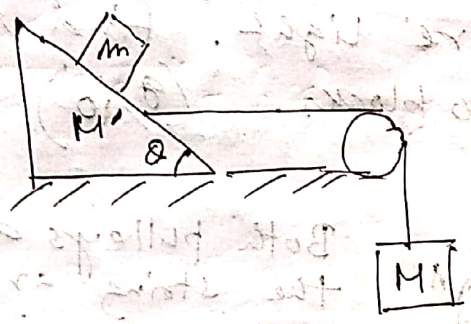
$\left[a_A = \frac{10g}{13} \text{ forward}, \; a_B = \frac{5g}{13} \text{ down}\right]$

c)



$\left[a_A = \frac{2g}{3} \text{ down}, \; a_B = \frac{g}{3} \text{ up}\right]$

6.



All surfaces are frictionless and the strings and pulleys are light. Find the mass 'M' of the hanging block in fig which will prevent the smaller block from slipping over the triangular block of mass 'M'.

$\left(\frac{M'+m}{\cot\theta - 1}\right)$

7. A car is speeding up on a horizontal road with an accl^n 'a'. A block is kept on a smooth incline inside the car and it does not slip on the incline. Find the angle of the incline with the horizontal.

$\left[\tan^{-1}(a/g)\right]$

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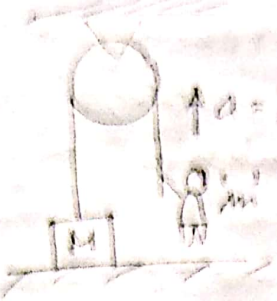
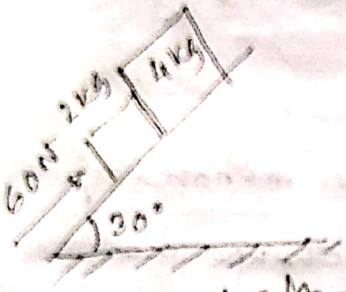
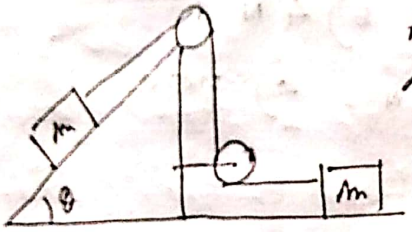
8. 

If fig the block of mass 'M' is at rest on the floor. The acceleration with which should a man boy of mass 'm' climb along the rope of negligible mass so as to lift the block from the floor is a) $\left(\frac{M}{m}-1\right)g$ b) $>\left(\frac{M}{m}-1\right)g$ c) $\frac{M}{m}g$ d) $>\frac{Mg}{m}$

9. 

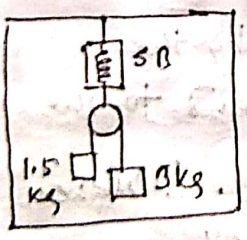
The inclined plane is frictionless. A force 60N is applied parallel to the plane in upward direction. Find normal contact force betn the two blocks.

a) 20N b) 40N c) 60N d) 80N

10. 

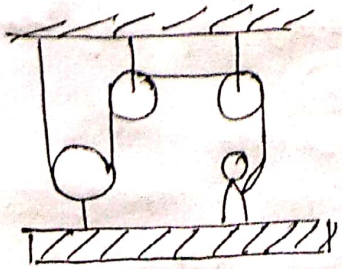
All surfaces are smooth, pulleys are light & frictionless. Tension in light string is

a) $\frac{2}{3}mg\sin\theta$ b) $\frac{3}{2}mg\sin\theta$ c) $\frac{mg\sin\theta}{2}$ d) $2mg\sin\theta$

11. 

$\uparrow \frac{g}{10}$

Find reading of spring balance in fig. The lift is going up with an accel of g/10. pulley & string are light & pulley is massless. [4.4 kg or 44 N]

12 

A 50 kg person stands on a 25 kg platform. The ~~system~~ platform moves up with constant velocity. Find the force with which the person pulls the rope? (250 N)

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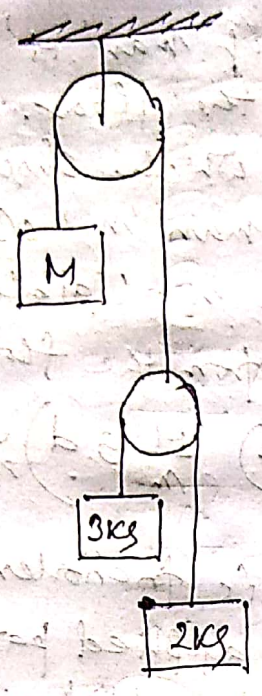
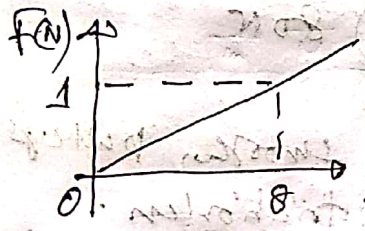
13. 

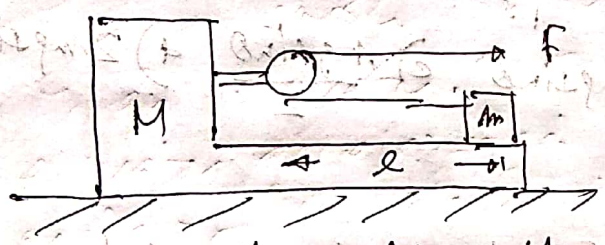
Find M so that it remains at rest. pulleys & strings are light & frictionless.

[4.8 kg]

14. A force – time graph for the motion of a body is shown in fig. change in linear momentum betn. 0 and 8 sec is



a) 0 b) 4 N-s
c) 8 N-s d) none

15. 

In the given fig. all surfaces are smooth. Find the time taken by the block (m) to reach from the free end to the pulley attached to the plank (M). Distance betn. free end & pulley is 'l'.

$$t = \sqrt{\frac{2Mml}{(2m+M)F}}$$

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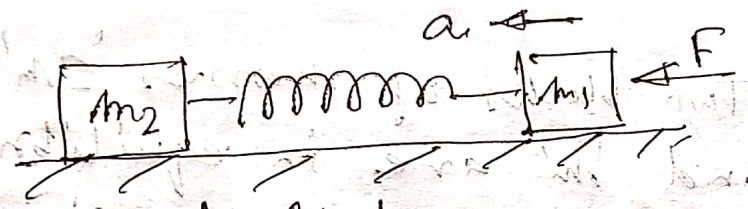

frictionless

Two blocks of masses m_1 & m_2 are connected to each other by a spring. If pushing force F is applied to mass m_1 & it gives accel^n 'a' to it, then find accel^n of m_2?

$$\left[\frac{F - m_1 a}{m_2}\right]$$

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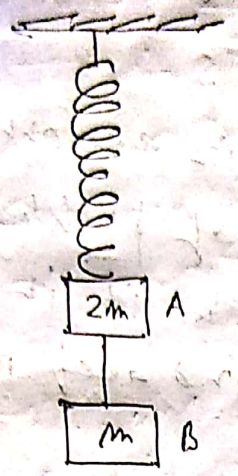
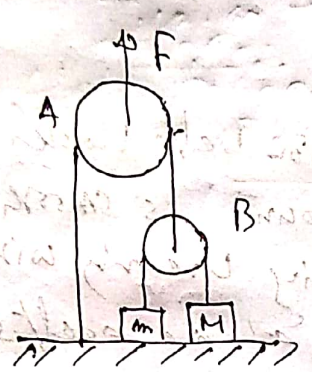
17. 

Two blocks of masses '2m' and 'm' are in equilibrium as in fig. Now the string between the blocks is suddenly broken. Find accl^n of the blocks A & B resp at that instant

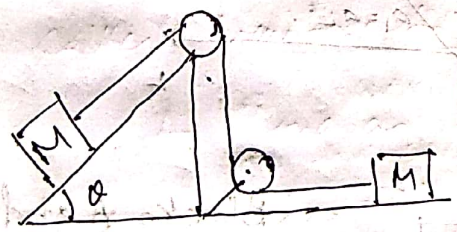
['g' downward, $\frac{g}{2}$ upward]

18. 

$m = 5kg$, $M = 10kg$, $F = 300N$. Find accl^n of 'm' & 'M'

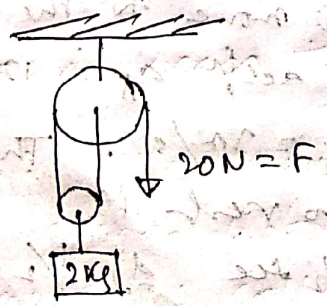
[$a_m = 5m/s^2$ up, $a_M = 0$]

19/ Two blocks of mass 5kg & 3kg are attached to the ends of a string passing over a smooth pulley fixed to the ceiling of an elevator. A man inside the elevator accelerated upward, finds the accl^n of the blocks to be $9\frac{1}{32}$. Find accl^n of elevator? [$a = \frac{9}{8}$]

20 

All surfaces & pulleys are frictionless. Find tension in string?

[$(Mg\sin\theta)/2$]

(21) 

Find accl^n of 2kg block. ($10m/s^2$)

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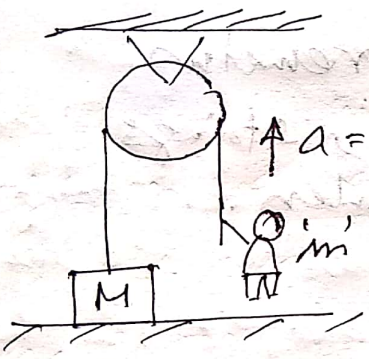
8.



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